

Examining the width of an introduction

The following introduction is taken from *Reducing food's environmental impacts through producers and consumers*, J. Poore and T. Nemecek, Science, 2018, 360, 6392, 987-992.

Task 1 – read the introduction

With current diets and production practices, feeding 7.6 billion people is degrading terrestrial and aquatic ecosystems, depleting water resources, and driving climate change ([1](#), [2](#)). It is particularly challenging to find solutions that are effective across the large and diverse range of producers that characterize the agricultural sector. More than 570 million farms produce in almost all the world's climates and soils ([3](#)), each using vastly different agronomic methods; average farm sizes vary from 0.5 ha in Bangladesh to 3000 ha in Australia ([3](#)); average mineral fertilizer use ranges from 1 kg of nitrogen per ha in Uganda to 300 kg in China ([4](#)); and although four crops provide half of the world's food calories ([4](#)), more than 2 million distinct varieties are recorded in seed vaults ([5](#)). Further, products range from minimally to heavily processed and packaged, with 17 of every 100 kg of food produced transported internationally, increasing to 50 kg for nuts and 56 kg for oils ([4](#)).

Previous studies have assessed aspects of this heterogeneity by using geospatial data sets ([6–8](#)), but global assessments using the inputs, outputs, and practices of actual producers have been limited by data. The recent rapid expansion of the life cycle assessment (LCA) literature is providing this information by surveying producers around the world. LCA then uses models to translate producer data into environmental impacts with sufficient accuracy for most decision-making ([9–11](#)).

To date, efforts to consolidate these data or build new large-scale data sets have covered greenhouse gas (GHG) emissions only ([8](#), [12](#), [13](#)), agriculture only ([13–16](#)), small numbers of products ([8](#), [14–16](#)), and predominantly Western European producers ([12–16](#)) and have not corrected for important methodological differences between LCAs ([12–16](#)). Here, we present a globally reconciled and methodologically harmonized database on the variation in food's multiple impacts. Our results show the need for far-reaching changes in how food's environmental impacts are managed and communicated.

Note all the references in this: the opening is basically a miniature literature review.

Task 2 – read the introduction again and notice how the width changes from start to finish

Very wide first sentence: a problem of interest and importance to almost everyone.

With current diets and production practices, feeding 7.6 billion people is degrading terrestrial and aquatic ecosystems, depleting water resources, and driving climate change ([1](#), [2](#)).

Narrower second sentence, though still wide: now we're focused on the idea of *solving the problem*. It is particularly challenging to find solutions that are effective across the large and diverse range of producers that characterize the agricultural sector.

Narrowing down further, giving examples to support the second sentence. More than 570 million farms produce in almost all the world's climates and soils ([3](#)), each using vastly different agronomic methods; average farm sizes vary from 0.5 ha in Bangladesh to 3000 ha in Australia ([3](#)); average mineral fertilizer use ranges from 1 kg of nitrogen per ha in Uganda to 300 kg in China ([4](#)); and although four crops provide half of the world's food calories ([4](#)), more than 2 million distinct varieties are recorded in seed vaults ([5](#)). Further, products range from minimally to heavily processed and packaged, with 17 of every 100 kg of food produced transported internationally, increasing to 50 kg for nuts and 56 kg for oils ([4](#)).

Narrowing down still further, to describe previous studies into the problem. Previous studies have assessed aspects of this heterogeneity by using geospatial data sets ([6–8](#)), but global assessments using the inputs, outputs, and practices of actual producers have been limited by data. The recent rapid expansion of the life cycle assessment (LCA) literature is providing this information by surveying producers around the world. LCA then uses models to translate producer data into environmental impacts with sufficient accuracy for most decision-making ([9–11](#)).

Finally, we arrive at the challenge, the very specific task that this paper undertakes. To date, efforts to consolidate these data or build new large-scale data sets have covered greenhouse gas (GHG) emissions only ([8](#), [12](#), [13](#)), agriculture only ([13–16](#)), small numbers of products ([8](#), [14–16](#)), and predominantly Western European producers ([12–16](#)) and have not corrected for important methodological differences between LCAs ([12–16](#)). Here, we present a globally reconciled and methodologically harmonized database on the variation in food's multiple impacts.

And this is the preview, which we'll discuss later. Our results show the need for far-reaching changes in how food's environmental impacts are managed and communicated.

Overall, you can see that there is a dramatic decrease in width from the first sentence to the challenge. You can also see that the decrease is gradual and has a clear flow: it doesn't just jump from wide to narrow.